Discretization and Matrix Formulation of the 1D Pennes Bioheat Equation in Spherical Coordinates

1. Governing Equation: Pennes' Bioheat Equation

$$\rho c \frac{\partial T(r,t)}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \rho_b c_b \omega_b (T_b - T) + Q_{\text{met}} + Q_{\text{ext}}(T) \tag{1}$$

where:

- ρ, c: tissue density and specific heat
- k: thermal conductivity
- ρ_b, c_b, ω_b: blood density, specific heat, and perfusion rate
- T_b: arterial blood temperature
- Q_{met}: metabolic heat generation
- $Q_{\text{ext}}(T)$: temperature-dependent external heat source

Expanding the radial derivative:

$$\rho c \frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial r^2} + \frac{2}{r}\frac{\partial T}{\partial r}\right) + \rho_b c_b \omega_b (T_b - T) + Q_{\text{met}} + Q_{\text{ext}}(T) \tag{2}$$

2. Initial and Boundary Conditions

- **Initial condition:** $T(r,0) = T_0$
- **Symmetry boundary at** $r = 0$**:** $\frac{\partial T}{\partial r}\big|_{r=0} = 0$
- **Convective boundary at** $r = R$**:**

$$-k\left.\frac{\partial T}{\partial r}\right|_{r=R} = h(T(R,t) - T_\infty)$$

3. Discretization: Implicit Method

Let $r_i = (i-1)\Delta r$ for $i = 1, 2, \ldots, N$ where $r_1 = 0$, $r_N = R$, and $t^n = n\Delta t$. Define:

$$T_i^n \approx T(r_i, t^n), \quad \theta = \frac{\rho c}{\Delta t}, \quad \beta = \frac{k}{\Delta r^2}, \quad \lambda = \rho_b c_b \omega_b, \quad \gamma_i = \frac{k}{r_i \Delta r}$$

Node $i = 1$ (Symmetry Boundary)

Using a ghost point: $T_0 = T_2$, the second derivative becomes:

$$\left.\frac{\partial^2 T}{\partial r^2}\right|_{i=1} \approx \frac{2(T_2 - T_1)}{\Delta r^2} = 2\beta(T_2 - T_1)$$

The discretized equation:

$$\theta(T_1^{n+1} - T_1^n) = 2\beta(T_2^{n+1} - T_1^{n+1}) + \lambda(T_b - T_1^{n+1}) + Q_{\text{met}} + Q_{\text{ext}}(T_1^n)$$

Nodes $i = 2, \ldots, N-1$ (Interior)

$$\begin{aligned}\theta(T_i^{n+1} - T_i^n) = \beta(T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}) + \gamma_i(T_{i+1}^{n+1} - T_{i-1}^{n+1}) \\ + \lambda(T_b - T_i^{n+1}) + Q_{\text{met}} + Q_{\text{ext}}(T_i^n)\end{aligned}$$

Node $i = N$ (Convective Boundary)

Using backward difference:

$$-k \cdot \frac{T_N^{n+1} - T_{N-1}^{n+1}}{\Delta r} = h(T_N^{n+1} - T_\infty) \Rightarrow (k + h\Delta r)T_N^{n+1} - kT_{N-1}^{n+1} = h\Delta r T_\infty$$

4. Matrix Form

We solve the linear system:

$$\mathbf{A}\,\mathbf{T}^{n+1} = \mathbf{b}^n$$

Temperature Vector

$$\mathbf{T}^{n+1} = \begin{bmatrix} T_1^{n+1} \\ T_2^{n+1} \\ \vdots \\ T_N^{n+1} \end{bmatrix}$$

Right-Hand Side Vector

$$b_i^n = \begin{cases} \theta T_1^n + \lambda T_b + Q_{\text{met}} + Q_{\text{ext}}(T_1^n), & i = 1 \\ \theta T_i^n + \lambda T_b + Q_{\text{met}} + Q_{\text{ext}}(T_i^n), & 2 \le i \le N-1 \\ h\Delta r T_\infty, & i = N \end{cases}$$

Coefficient Matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$

$$\mathbf{A} = \begin{bmatrix} b_1 & c_1 & 0 & \cdots & 0 \\ a_2 & b_2 & c_2 & \cdots & 0 \\ 0 & a_3 & b_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & c_{N-1} \\ 0 & \cdots & 0 & a_N & b_N \end{bmatrix}$$

Matrix Coefficients

- For $i = 1$:

$$b_1 = \theta + 2\beta + \lambda, \quad c_1 = -2\beta$$

- For $2 \leq i \leq N - 1$:

$$a_i = -(\beta - \gamma_i), \quad b_i = \theta + 2\beta + \lambda, \quad c_i = -(\beta + \gamma_i)$$

- For $i = N$:

$$a_N = -k, \quad b_N = k + h\Delta r$$

5. Final Linear System

$$\mathbf{A}\mathbf{T}^{n+1} = \begin{bmatrix} \theta T_1^n + \lambda T_b + Q_{\text{met}} + Q_{\text{ext}}(T_1^n) \\ \theta T_2^n + \lambda T_b + Q_{\text{met}} + Q_{\text{ext}}(T_2^n) \\ \vdots \\ h\Delta r T_\infty \end{bmatrix}$$

Solve the linear system of equations:

$$\mathbf{A}\mathbf{T} = \mathbf{b}$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$, and $\mathbf{T} \in \mathbb{R}^n$ is the temperature vector.

6. Richardson Iteration Method

Iteration Formula

$$\mathbf{T}^{(n+1)} = \mathbf{T}^{(n)} + \omega \left(\mathbf{b} - \mathbf{A}\mathbf{T}^{(n)} \right)$$

where:

- $\mathbf{T}^{(n)}$ is the approximation at iteration step n
- $\omega > 0$ is the relaxation parameter
- $\mathbf{r}^{(n)} = \mathbf{b} - \mathbf{A}\mathbf{T}^{(n)}$ is the residual

Convergence Condition

The Richardson iteration converges if the relaxation parameter ω satisfies:

$$0 < \omega < \frac{2}{\rho(\mathbf{A})}$$

where $\rho(\mathbf{A})$ is the spectral radius of $\mathbf{A}$, i.e., the largest absolute value among its eigenvalues.

Algorithm

Inputs

- $\mathbf{A} \in \mathbb{R}^{n \times n}$: coefficient matrix
- $\mathbf{b} \in \mathbb{R}^n$: right-hand side vector
- $\mathbf{T}^{(0)} \in \mathbb{R}^n$: initial guess
- $\omega > 0$: relaxation parameter
- $\varepsilon > 0$: convergence tolerance

Procedure

1. Initialize $n = 0$, $\mathbf{T}^{(0)}$
2. Compute residual $\mathbf{r}^{(0)} = \mathbf{b} - \mathbf{A}\mathbf{T}^{(0)}$
3. While $\|\mathbf{r}^{(n)}\| \geq \varepsilon$, repeat:

$$\begin{aligned} \mathbf{T}^{(n+1)} &= \mathbf{T}^{(n)} + \omega \mathbf{r}^{(n)} \\ \mathbf{r}^{(n+1)} &= \mathbf{b} - \mathbf{A}\mathbf{T}^{(n+1)} \\ n &\leftarrow n + 1 \end{aligned}$$

4. Return $\mathbf{T}^{(n+1)}$ as the solution